

HAT-P-19b

R-1530

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_220826 · created 2026-08-01T18:02:49+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459817.892 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.054
Transit depth (Rp/Rs)^2	3.35 +/- 1.99 [%]
Orbital Inclination (inc)	89.81 +/- 1.56
Airmass coefficient 1 (a1)	2.03 +/- 0.29
Airmass coefficient 2 (a2)	-0.69 +/- 0.32
Scatter in the residuals of the lightcurve fit is	13.62 %
Best Comparison Star	#3 - [438, 211]
Optimal Aperture	2.72
Optimal Annulus	7.83
Transit Duration (day)	0.119 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.054 vs 0.1418 (deviation 0.73σ)
Duration fit vs published	0.119 d vs 0.1196 d (deviation 0.05σ)
Mid-time O–C	2.0 min
Depth SNR	3.2
Residual scatter	13.62 %

Light curve

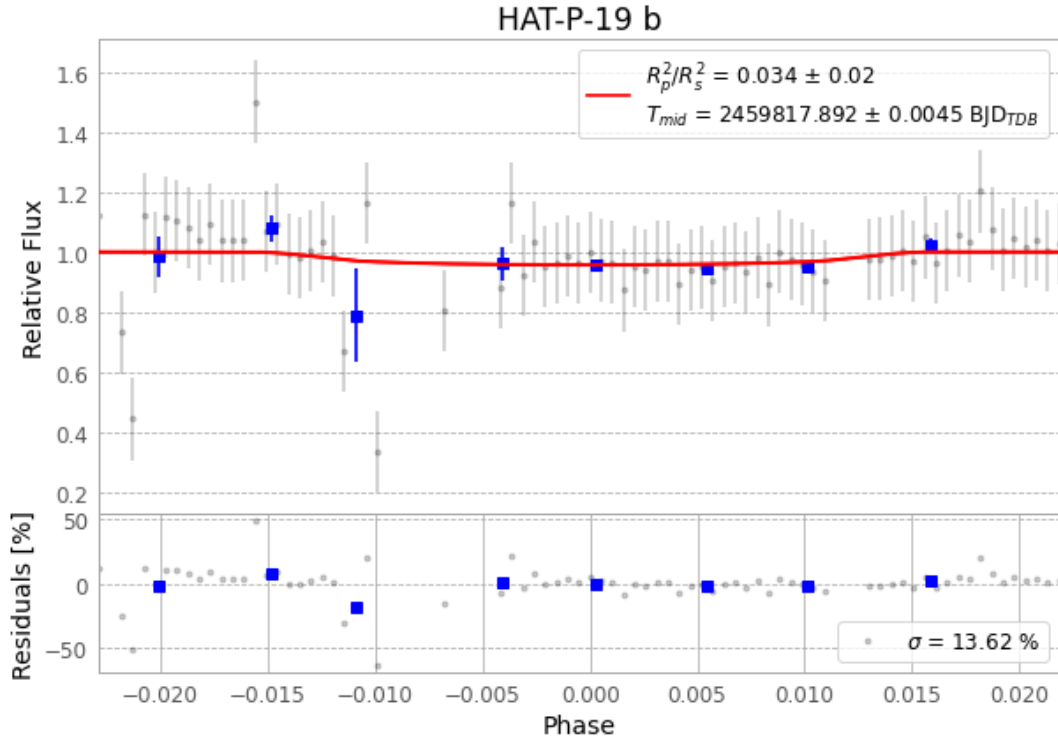


Figure 1 — FinalLightCurve_HAT-P-19 b_2022-08-26.png (SHA-256 43fbae69eb0144fa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [467, 238], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b86e62a2587979bc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-19220826070613.FITS → HATP-19220826112413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-220826.log (dbcb2636adf...), FinalLightCurve_HAT-P-19 b_2022-08-26.png (43fbae69eb01...), FinalLightCurve_HAT-P-19 b_2022-08-26.csv (905453b689d9...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 18:02Z.